

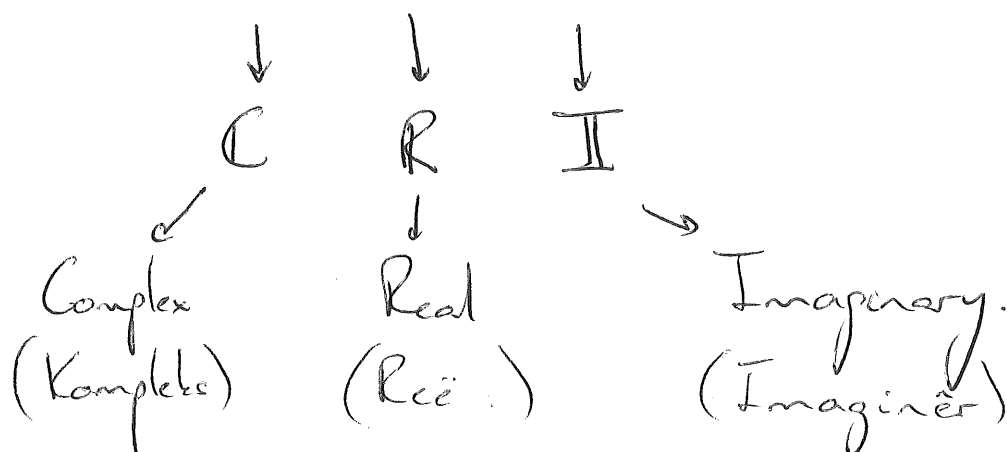
Appendix B

①

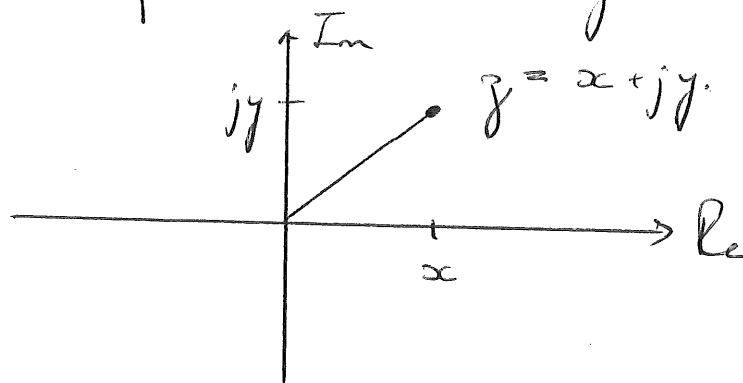
* Komplekse getalle.

$$\rightarrow z = x + jy$$

$$\boxed{j = \sqrt{-1}}$$



- Elke komplekse getal bestaan uit 'n reële en 'n imaginêre gedeelte. Hierdie kan in die komplekse vlak aangedui word:



- Komplekse getalle kan op 3 maniere voorgestel word.

$$z = x + jy$$

$$= r \angle \theta$$

$$= r e^{j\theta}$$

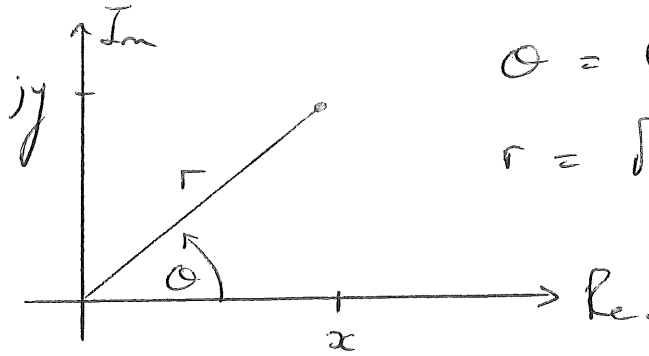
(Reghoekig)

(Polêr)

(Eksponensieel)

* Reghoekig, Polêr en Eksponensieel.

(2)



$$\theta = \tan^{-1}(y/x)$$

$$r = \sqrt{x^2 + y^2}$$

① Reghoekig $z = x + jy$
 $= r \cos \theta + j r \sin \theta$

② Polêr $z = r \angle \theta$

③ Eksponensieel $z = r e^{j\theta}$
waar $e^{j\theta} = \cos \theta + j \sin \theta$
 $= 1 \angle \theta$

Onthou

$$1 \angle \phi = e^{j\phi} = \cos \phi + j \sin \phi$$

$$1 \angle -\phi = e^{-j\phi} = \cos(-\phi) + j \sin(-\phi)$$
$$= \cos \phi - j \sin \phi$$

$\therefore 1 \angle \pm \phi = e^{\pm j\phi} = \cos \phi \pm j \sin \phi.$

↳ Euler se Wet!

* Wiskundige formules

③

① Sammening (Reghoekig)

$$\begin{aligned} z_1 + z_2 &= (x_1 + jy_1) + (x_2 + jy_2) \\ &= (x_1 + x_2) + j(y_1 + y_2). \end{aligned}$$

② Aftræk (Reghoekig)

$$\begin{aligned} z_1 - z_2 &= (x_1 + jy_1) - (x_2 + jy_2) \\ &= (x_1 - x_2) + j(y_1 - y_2). \end{aligned}$$

③ Vermenigvuldiging. (Reghoekig of Polêr)

$$\begin{aligned} z_1 \times z_2 &= (x_1 + jy_1)(x_2 + jy_2) \\ &= x_1 x_2 + jy_1 x_2 + jy_2 x_1 + j^2 y_1 y_2 \\ &= x_1 x_2 - y_1 y_2 + j(y_1 x_2 + y_2 x_1). \end{aligned}$$

$$\begin{aligned} \text{of } z_1 \times z_2 &= r_1 \angle \theta_1 \times r_2 \angle \theta_2 \\ &= r_1 e^{j\theta_1} \times r_2 e^{j\theta_2} \\ &= r_1 r_2 e^{j(\theta_1 + \theta_2)} \\ &= r_1 r_2 \angle (\theta_1 + \theta_2) \end{aligned}$$

④ Deling (Polêr)

④

$$\begin{aligned}\frac{r_1 \angle \theta_1}{r_2 \angle \theta_2} &= \frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} \\ &= \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)} \\ &= \frac{r_1}{r_2} \angle \theta_1 - \theta_2\end{aligned}$$

⑤ Vierkantwortel (Polêr)

$$\begin{aligned}\sqrt{z} &= \sqrt{r \angle \theta} = \sqrt{r e^{j\theta}} \\ &= (r e^{j\theta})^{\frac{1}{2}} \\ &= \sqrt{r} e^{j\theta/2} = \sqrt{r} \angle \theta/2.\end{aligned}$$

⑥ Inverse (Polêr)

$$\frac{1}{z} = \frac{1 \angle 0^\circ}{r \angle \theta} = \frac{1}{r} \angle 0 - \theta = \frac{1}{r} \angle -\theta$$

⑦ Komplekse toegevoegde

$$\begin{aligned}z^* &= (x + jy)^* \\ &= x - jy\end{aligned}$$

$$\begin{aligned}z^* &= (r \angle \theta)^* \\ &= r \angle -\theta\end{aligned}$$

Voorbeelde

(5)

① Teken die volgende ~~getalle~~ op die komplekse vlak en skryf in polêre formaat.

• $j = \dots \angle \dots$

• $-j = \dots \angle \dots$

• $-1 =$

• $+1 =$

• $1+j =$

• $1-j =$

• $-1+j =$

• $-1-j =$

② Bereken:

$$[(s+j2)(-1+j4) - 5\angle 60^\circ]^* = ?$$

③ Bereken:

$$\frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} = ?$$

④ Bereken:

$$\frac{\sqrt{2}(-6+j4)(5-j)}{-1+j\sqrt{3}} = ?$$